

DETAILED DESCRIPTION OF THE INVENTION

Definitions and Terminology Clarification

For purposes of the present disclosure, the term “penile anatomical geometry profile” refers to a structured digital representation comprising spatially indexed parameters describing at least longitudinal length and radial surface distribution of penile anatomy, whether expressed as raw measurements, reconstructed surface models, or derived structural metrics.

The term “radial distance measurement” refers to a depth value representing the distance between a sensing reference plane and a point on an anatomical surface, regardless of the specific sensing modality used to acquire such value.

The term “longitudinal displacement” refers to positional indexing data corresponding to movement of a sensing device relative to an anatomical axis, whether determined by optical motion sensing, inertial sensing, mechanical encoding, magnetic tracking, or other displacement detection techniques.

The term “time-of-flight sensor” encompasses direct ToF, indirect ToF, structured light, lidar, phase-shift ranging, pulsed optical depth sensing, multi-zone depth arrays, and any equivalent depth acquisition technology capable of resolving spatial distance values.

The term “optical motion sensing” encompasses optical flow sensors, speckle-based displacement detectors, emitter-receiver displacement tracking systems, camera-based motion estimation modules, or other systems capable of detecting relative movement across a surface.

The term “geometry reconstruction engine” refers to any computational architecture configured to transform synchronized radial distance and positional data into a structured surface representation or derived geometric metrics.

The term “encoded geometry profile” includes any compressed, encrypted, normalized, vectorized, tokenized, hashed, or otherwise transformed data representation preserving structural characteristics derived from measured anatomy.

The term “compatibility framework” refers to any computational system configured to compare two or more anatomical geometry profiles to generate structural alignment metrics, similarity scores, proportionality evaluations, or matching outputs.

Unless otherwise expressly limited, terminology used herein shall be interpreted broadly to encompass functionally equivalent technologies, future-developed sensing modalities, alternative computational techniques, and structurally similar embodiments capable of performing the disclosed synchronized spatial acquisition and geometry profiling processes.